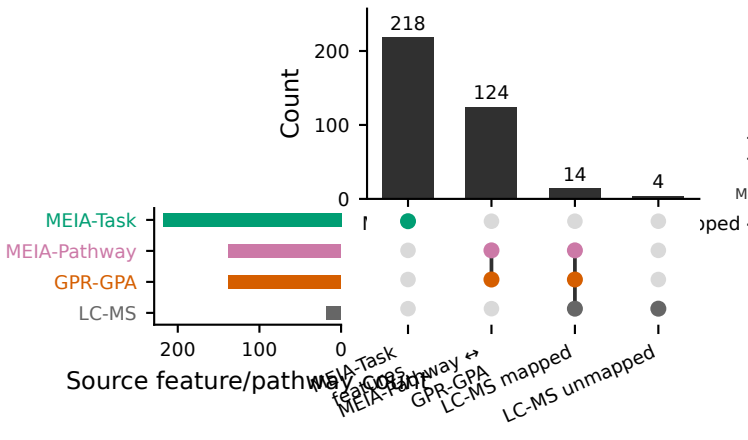
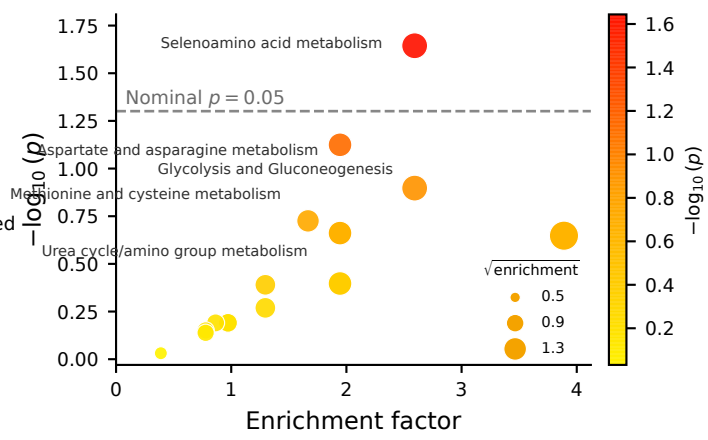
A Treatment four-set metabolic overlap

LC-MS: 22 pathways → 18 harmonized
(14 mapped + 4 unmapped); MEIA: 218 tasks + 138 pathways.



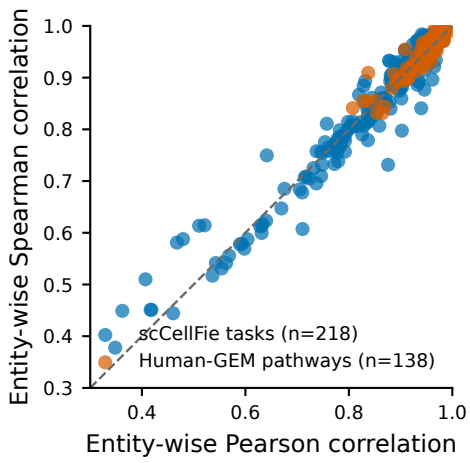
B Treatment-associated pathway enrichment

22 candidates shared with A; top five labeled.
3 pass pathway BH-FDR < 0.05.



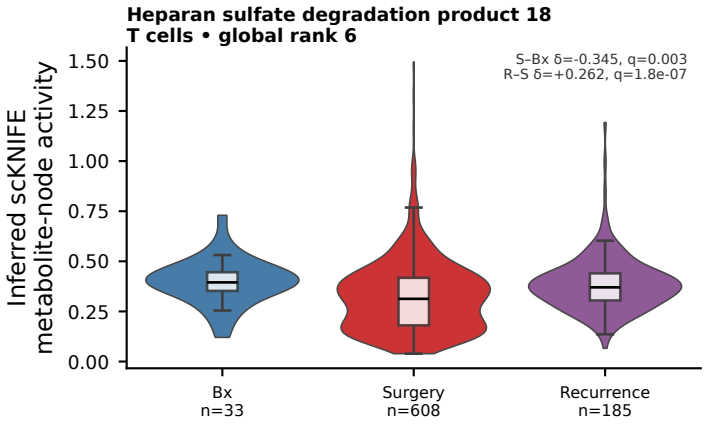
C MEIA preserves matched scKNIFE activity

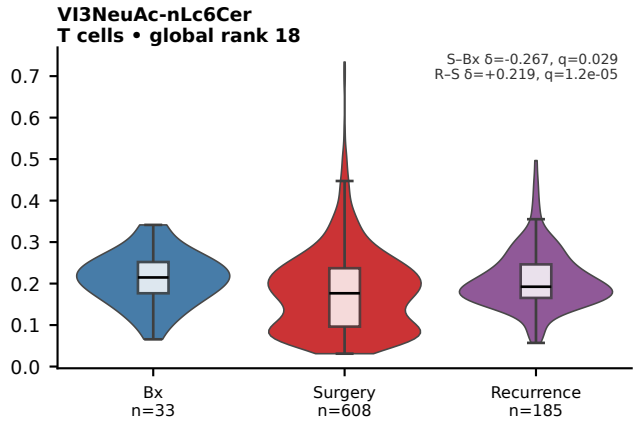
356 entities (218 tasks + 138 pathways); each entity correlated across 69,39

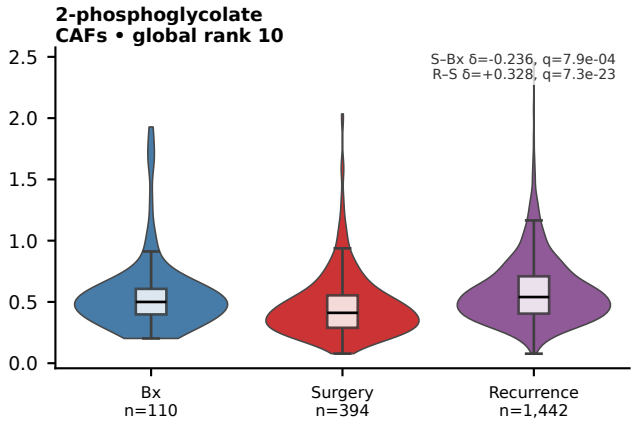


D Mechanism-aligned longitudinal restoration

P3: Bx → Surgery → Recurrence; strict Tier-3 inferred metabolite nodes.



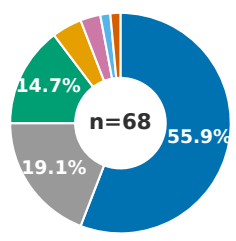




Cell-level exploratory distributions from one recurrence patient; the six-patient cohort independently supplies Bx–Surgery discovery evidence

E Tier-3 pathway context

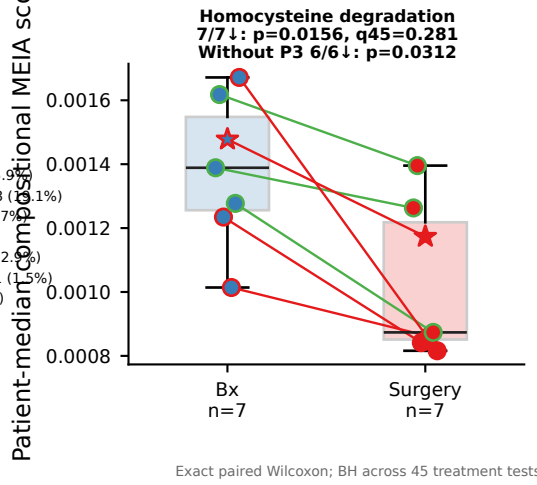
68 unique inferred metabolite nodes; each counted once.



context; generic exchange, transport, isolated and miscellaneous groups excluded.

F Macrophage metabolic tasks

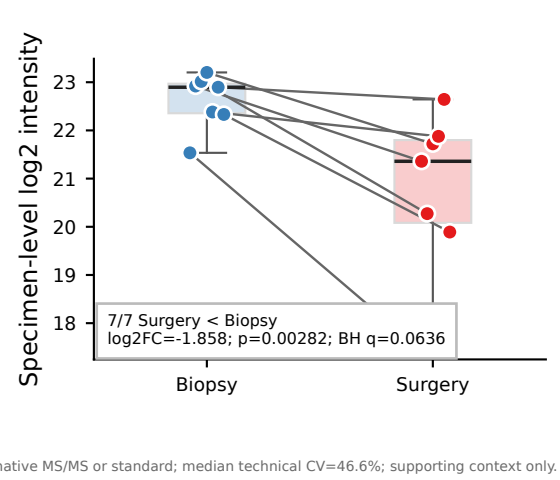
Paired Bx–Surgery; seven patients plus leave-P3-out sensitivity.



Exact paired Wilcoxon; BH across 45 treatment tests. Star=P3; line color=metastatic status. Neither task passes q<0.05.

G Paired LC–MS m/z 215.0260

Accurate-mass candidate: Se-adenosyl-L-selenohomocysteine



No native MS/MS or standard; median technical CV=46.6%; supporting context only.

H NG8 entity subgraph of the scKNIFE knowledge backbone

20 NG8 nodes plus 15 minimum connectors; topology is biochemical context, not node-level statistical evidence.

